



General histology

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List of Contents:

- Learning outcome
- Specialized CT Cartilage.
- Functions of cartilage.
- Types of cartilage:

Learning outcome:

- Student must have a clear understanding of each type of proper CT & specialized CT (cartilage).
- Student has to know both different types of cartilage and how to distinguish both type, location and function of each of them.

Specialized Connective tissue

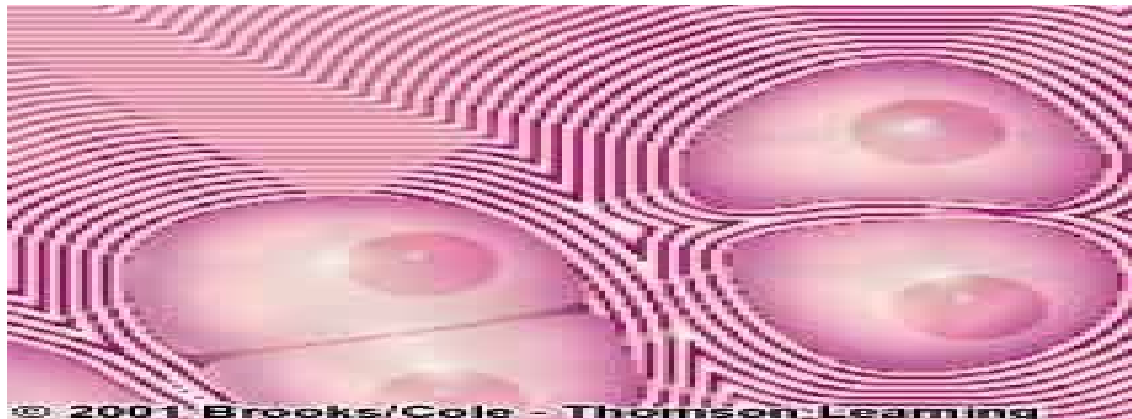
Cartilage

- Cartilage is a specialized form of connective tissue. It forms most of the fetal skeleton & persists in sites, most fetal cartilage eventually becomes bone.
- **Functions of cartilage are:**
 - 1- Its bear mechanical stresses without permanent distortion.
 - 2-support soft tissues.
 - 3-permits the elongation of long bones

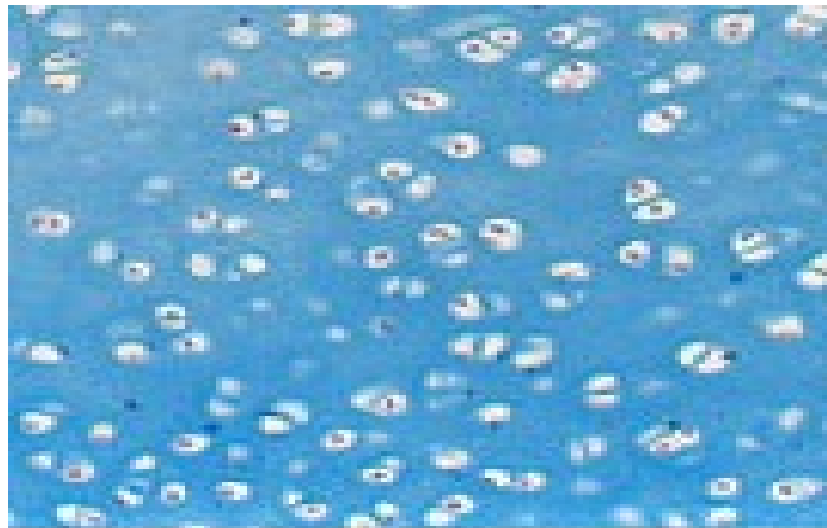
Like other all connective tissues, cartilage is composed of

- 1- Cells called chondrocytes.
- 2- Fibers.
- 3- Ground substance.
- Most cartilage types are enveloped by a layer of dense connective tissue called the perichondrium which contains the blood supply also contains nerves and lymphatic vessels and fibroblastic stem cells from which additional chondrocytes may arise.

so the ground substance composition controls the transport of nutrients & oxygen to chondrocytes from the surrounding vessels. Under light microscope, chondrocytes appear rounded, with an eccentric nucleus, and basophilic cytoplasm, the chondrocytes .contains well developed RER & Golgi complex



- Chondrocytes synthesize the fibers (collagen and elastin in elastic cartilage)& Ground substance (hyaluronic acid, glycoprotein and proteoglycan). The cells are located in matrix cavities called lacuna.



There are 3 types of cartilage:



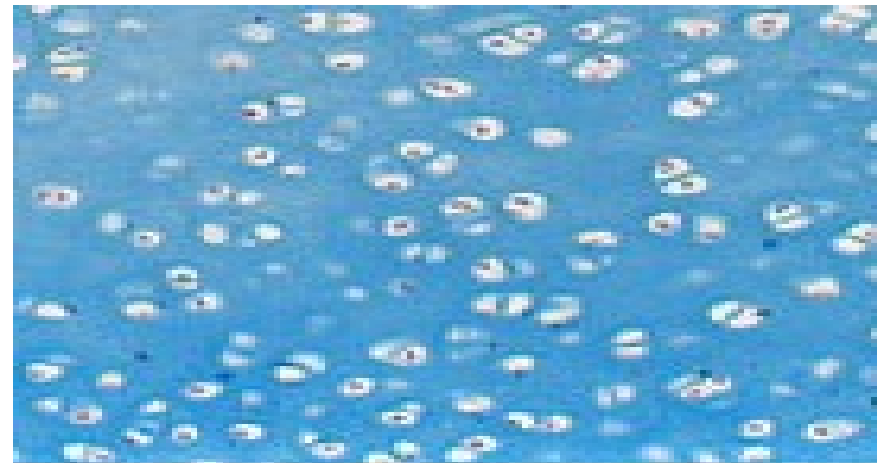
- 1- hyaline cartilage
- 2- elastic cartilage
- 3- fibro cartilage

1- Hyaline Cartilage

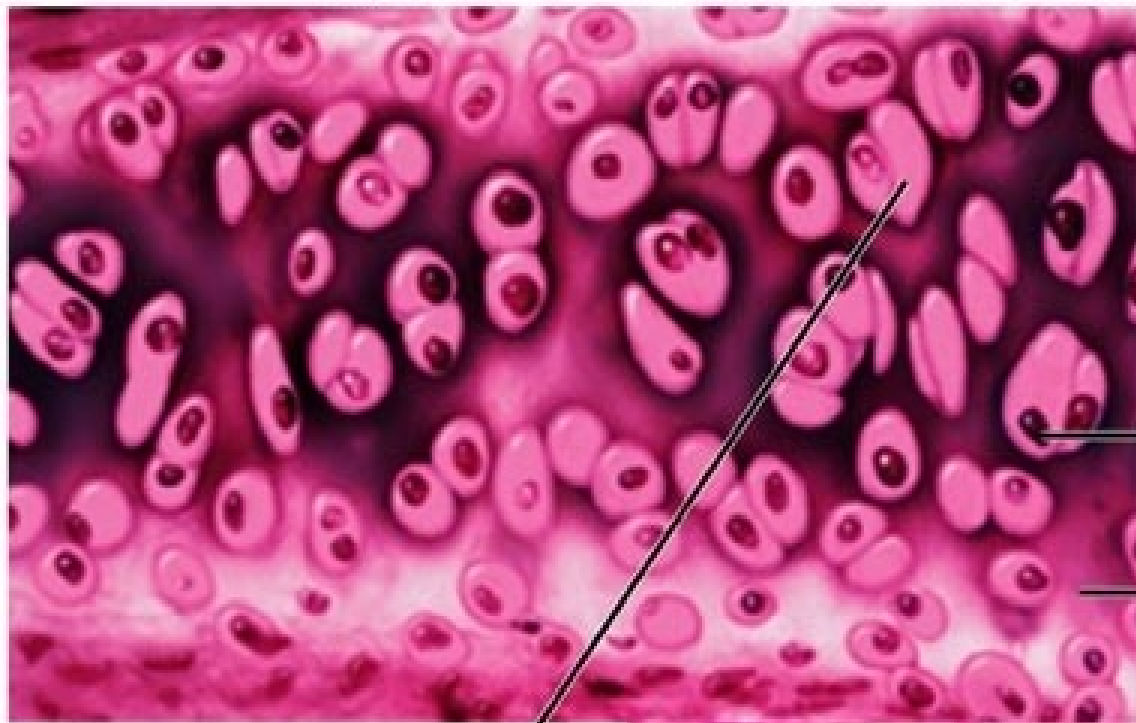
- Is the most common type in both fetus & adult, is white & translucent when fresh with a firm, gel like consistency.
- Hyaline cartilage composed of :
- **1- Fibers:**
- Hyaline cartilage matrix contains thin fibrils of type II collagen, which cannot be seen within the light microscope because of their small size.

2- Ground substance

- Composed of glycoprotein and proteoglycans
- the matrix which immediately surrounding the chondrocytes called the capsular or territorial matrix, & is more basophilic than the rest of the matrix which called the interterritorial (intercapsular) matrix.



Hyaline Cartilage



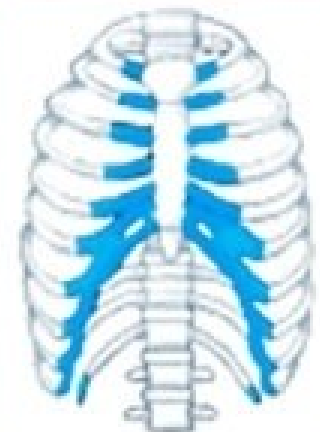
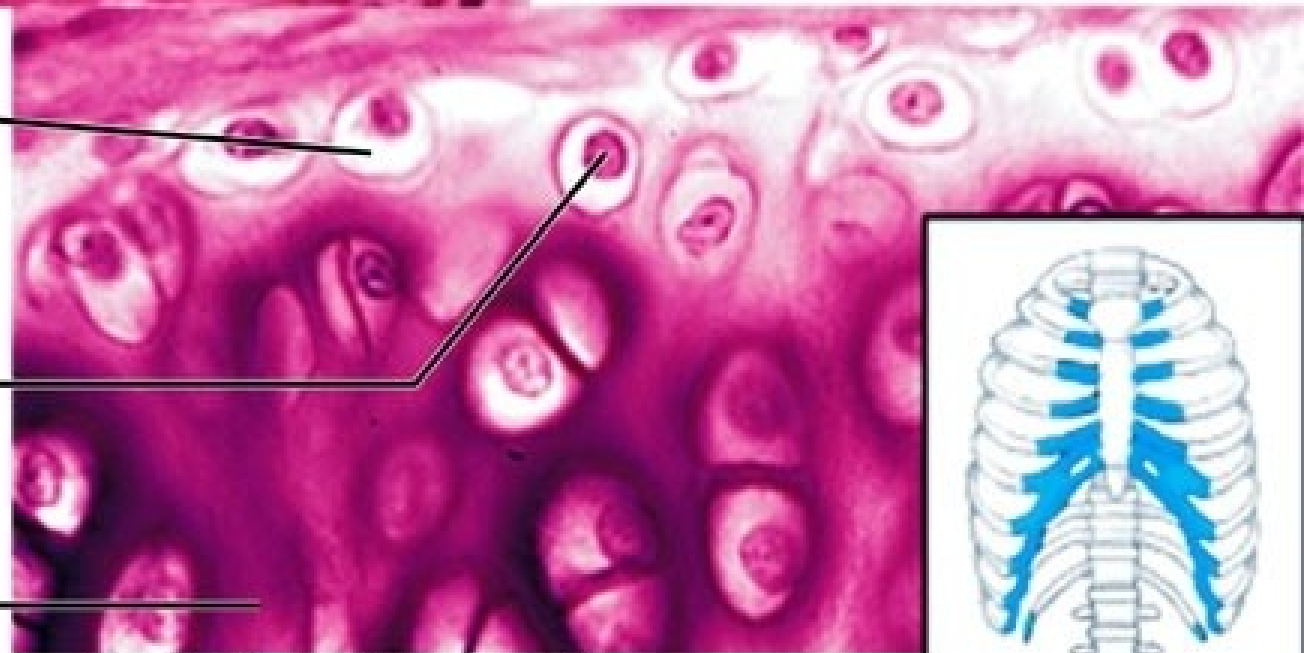
Nucleus

Intercellular material

Chondrocyte
in lacuna

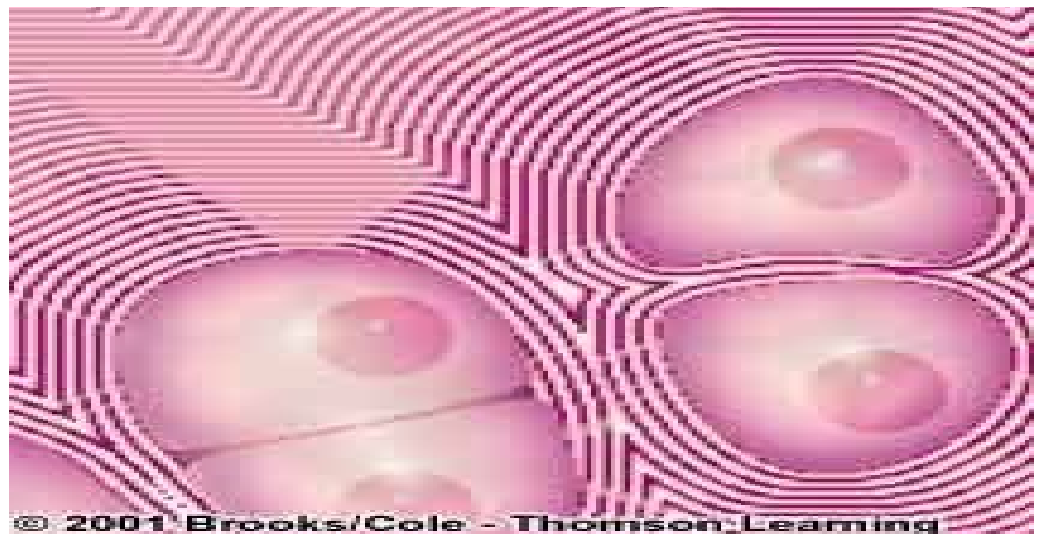
Nucleus

Intercellular
material



3- Cells (Chondrocytes)

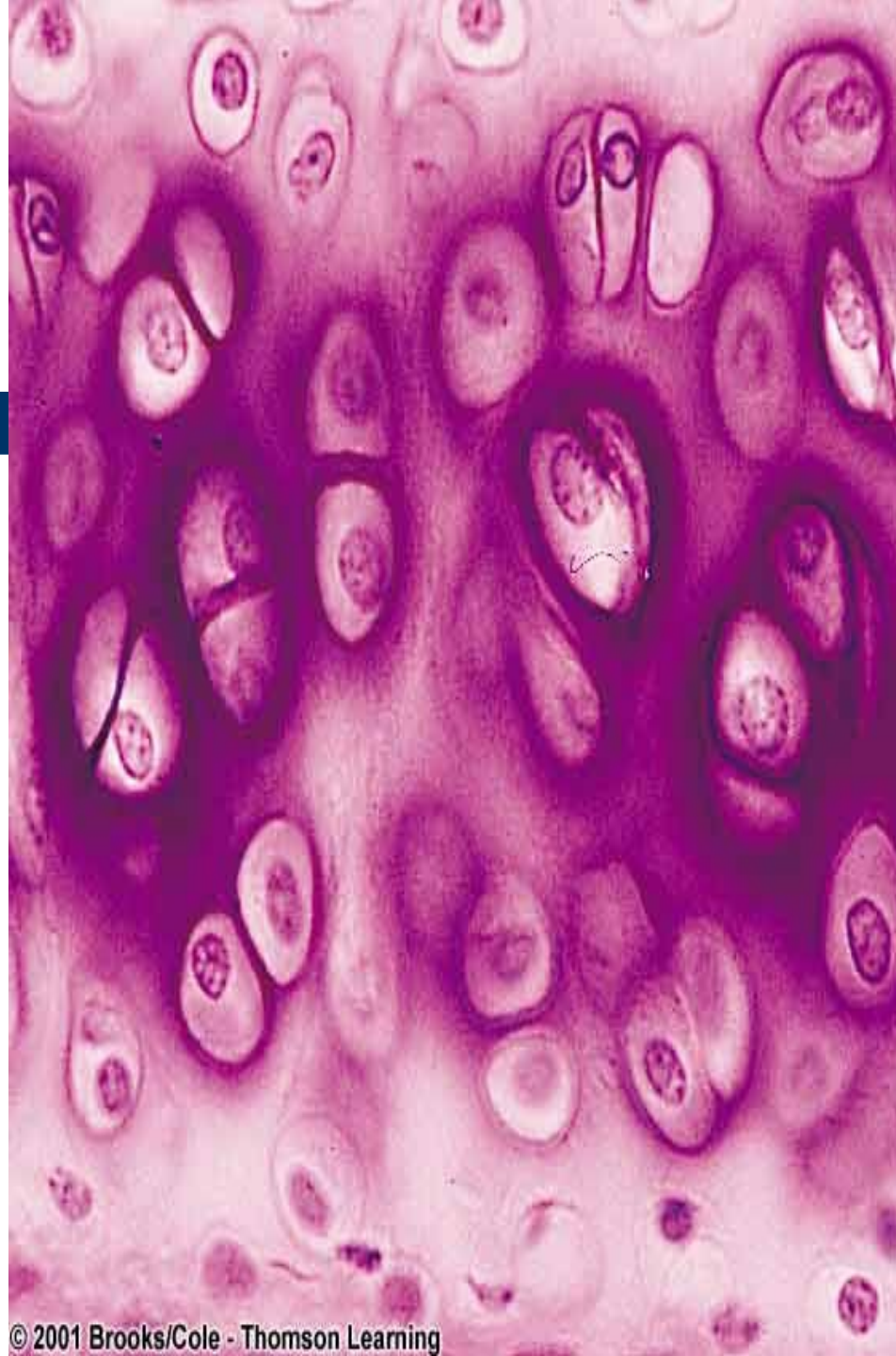
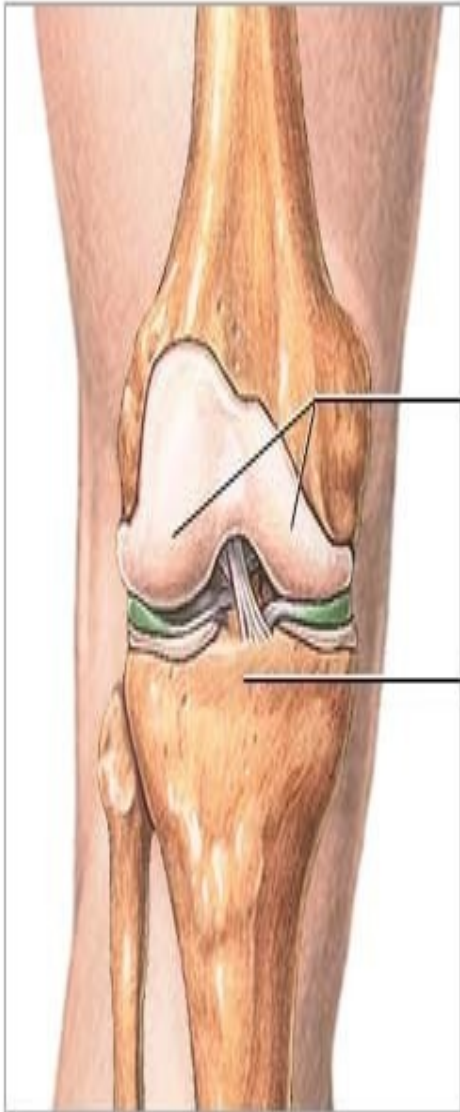
- Found embedded in the matrix either single or in isogenous groups of two to eight cells derived from one parent cell .
- The potential space occupied by each chondrocytes called a lacuna .



4- Perichondrium:

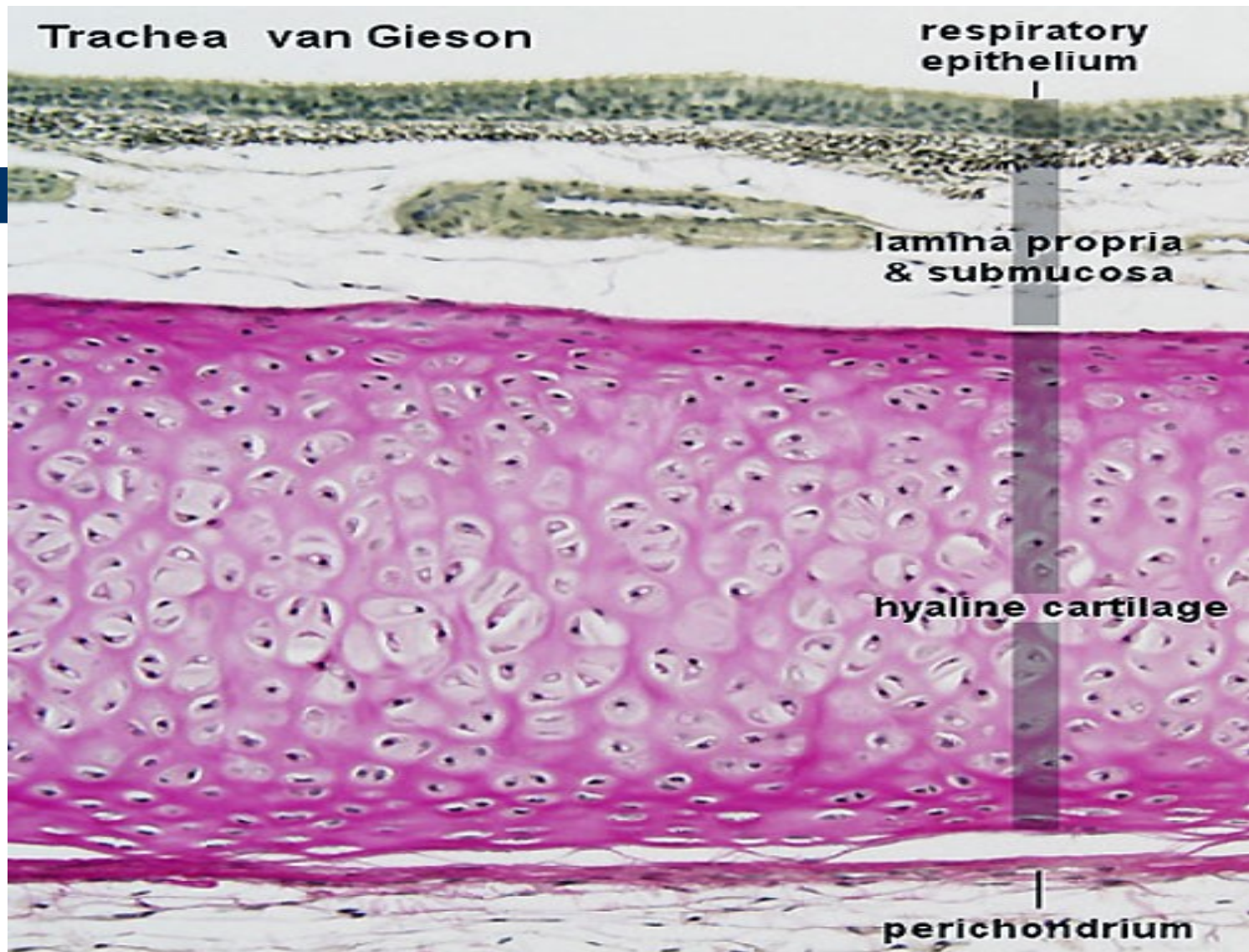
All hyaline cartilage surrounded and nourished by perichondrium except the articular cartilage of joints is nourished by synovial fluid.

- **location:**
- In adult mammals is found in:
 - - articular surfaces
 - - in large respiratory passages(nose, larynx, trachea, bronchi)
 - - in ventral ends of ribs.



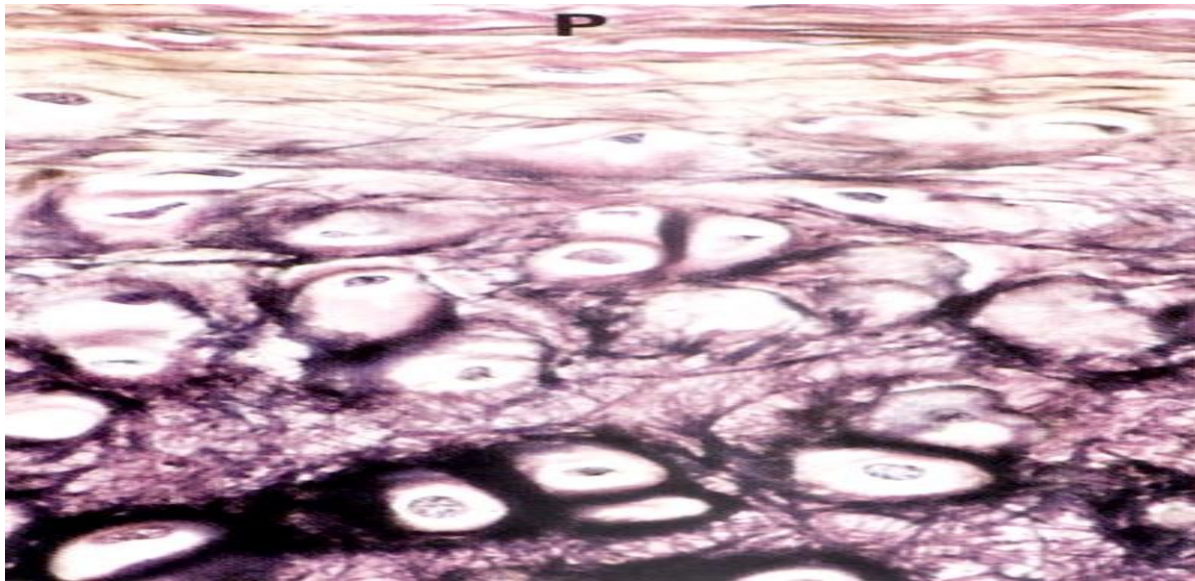
articular cartilage of joints

Trachea van Gieson

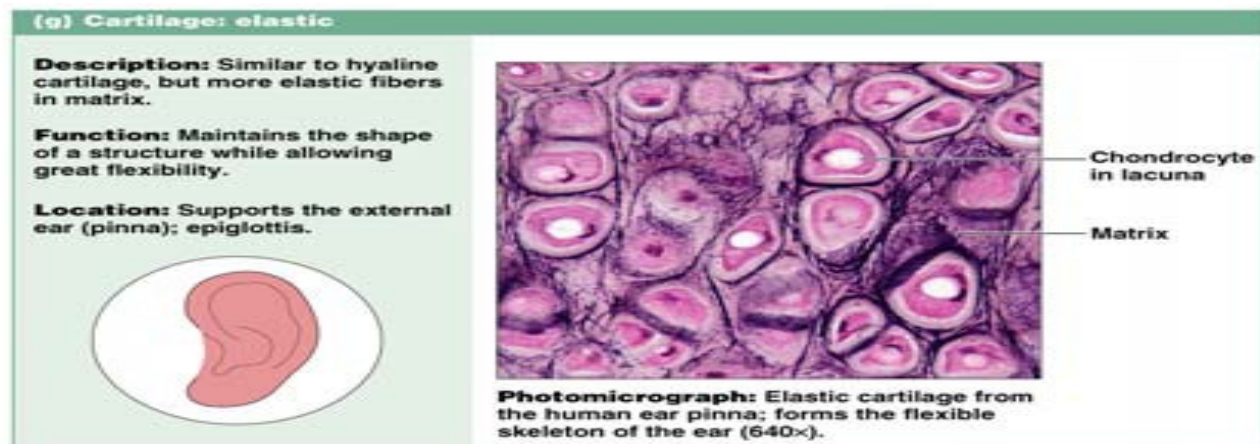


2-Elastic Cartilage

- Is yellowish when fresh and more flexible than hyaline.
- Fibers are resemble the hyaline cartilage but contains in addition of type II collagen, a dense network of branching elastic fibers.



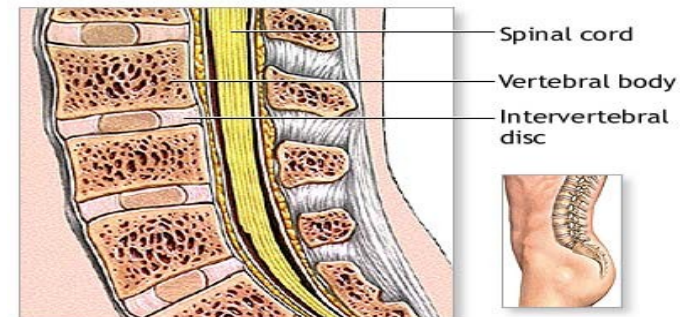
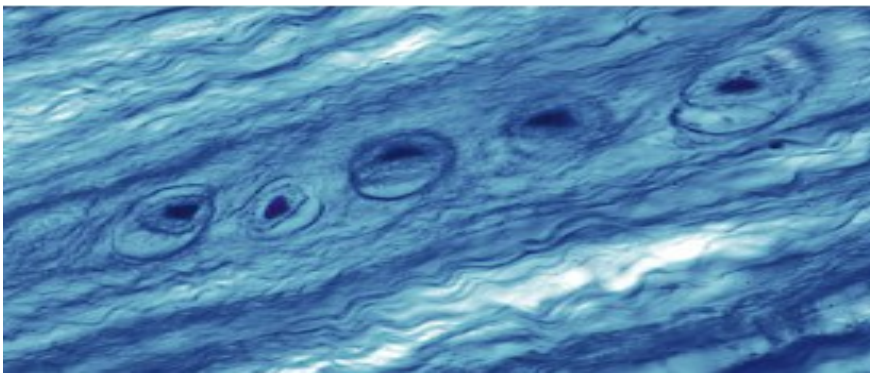
- The cells and ground substance is similar to that of hyaline cartilage.
- Elastic cartilage is surrounded by perichondrium.
- -location : is found in auricle of ear, external auditory canals , Eustachian tube , the epiglottis & in larynx.



3-Fibro Cartilage

Is intermediate in character between hyaline cartilage & dense connective tissue.

It contains abundant type I collagen fibers with some collagen type II .The fibers present as densely packed bundles and the chondrocytes are either singly or in isogenous groups usually arranged in long rows separated by coarse collagen type I fibers.



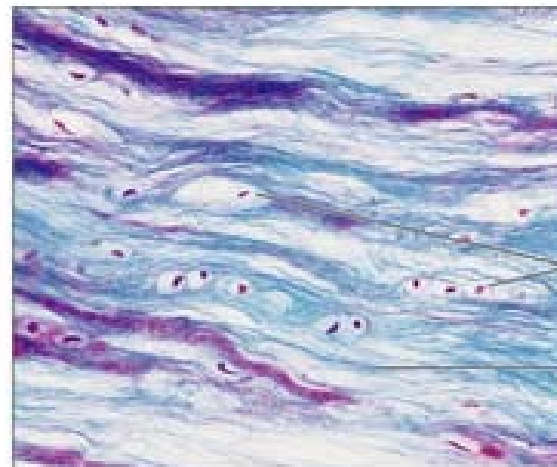
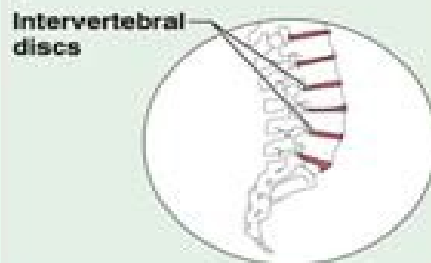
- there is no perichondrium.
- - location : is found in Intervertebral disks, discs of knee joint)

(h) Cartilage: fibrocartilage

Description: Matrix similar to but less firm than that in hyaline cartilage; thick collagen fibers predominate.

Function: Tensile strength with the ability to absorb compressive shock.

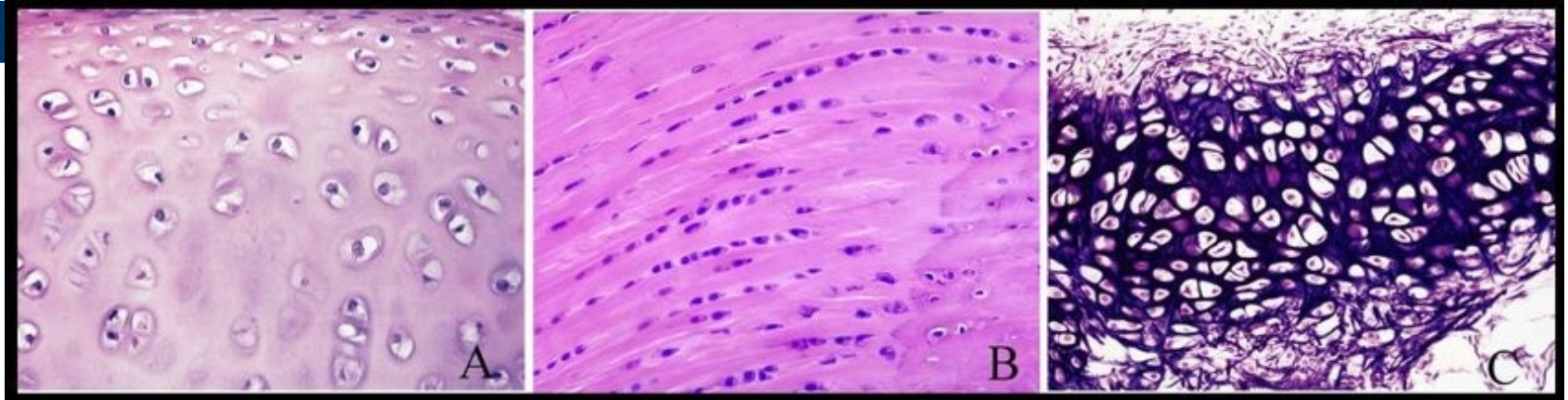
Location: Intervertebral discs; pubic symphysis; discs of knee joint.



Chondrocytes
in lacunae

Collagen
fiber

Photomicrograph: Fibrocartilage of an intervertebral disc (200×). Special staining produced the blue color seen.



Different types of cartilage. A) Hyaline cartilage; B) fibrocartilage; C) elastic cartilage.

A decorative graphic consisting of a light green L-shaped bar in the top-left corner and a dark blue horizontal bar spanning across the top of the slide.

Thanks